

Modeling Static and Dynamic Dependence Between Cryptocurrency and Traditional Financial Markets: Evidence from Vine and Time-Varying Copulas

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Abstract

Vine copula modelling and Bai-Perron structural break tests are used to demonstrate significant asymmetric tail dependence and regime shifts in the dependence structure between Bitcoin, equity, and currency markets.

This study investigates the static and dynamic dependence structure between Bitcoin (BTC), the S&P 500 and the US Dollar Index (DXY) over the period January 2021–December 2025. Marginal distributions are filtered using optimally selected GARCH-type models with Student- t and skewed Student- t innovations. Dependence is modeled using a static C-vine copula, a dynamic conditional correlation-GARCH (DCC-GARCH) model, and a rolling dynamic C-vine copula. Models are compared using out-of-sample predictive log-likelihoods and Diebold–Mariano tests.

The results reveal asymmetric dependence structures, including upper tail dependence between Bitcoin and the S&P 500 and negative dependence between BTC and the DXY index conditional on equity market movements. Out-of-sample evaluation shows that the static C-vine copula provides the best predictive performance. Structural break tests further identify several regime shifts associated with major events in the cryptocurrency market, most notably the FTX collapse, which is recovered automatically as a break in BTC–DXY dependence. Overall, the results indicate that BTC exhibits significant co-movement with equity markets, limiting its role as a diversification asset. This co-movement also suggests that stress can spread from cryptocurrency markets into traditional ones, a risk that grows as digital assets become more embedded in mainstream portfolios.

Keywords: Bitcoin; vine copula; tail dependence; DCC-GARCH; structural breaks; cryptocurrency; predictive evaluation

JEL Classification: C58, G11, G15

1 Introduction

Measuring dependence between financial assets is central to portfolio construction, risk management, and contagion analysis. Yet standard linear correlation misses the asymmetries and tail clustering that matter most in stress periods [Embrechts et al. \[2002\]](#),
35 [Longin and Solnik \[2001\]](#). Vine copulas address this limitation by decomposing the multivariate density into a cascade of bivariate pair-copulas, each capturing a distinct tail and asymmetry profile [Joe \[1996\]](#), [Bedford and Cooke \[2002\]](#), [Aas et al. \[2009\]](#). Applied to cryptocurrency markets, however, two questions remain open: whether static or dynamic specifications yield superior out-of-sample predictions, and whether the dependence
40 structure itself is stable or subject to structural breaks.

These two questions matter beyond methodology. Cryptocurrency is no longer a side bet confined to retail traders: banks hold crypto exposure, ETFs tie BTC directly to institutional portfolios, and a shock in one market no longer stays contained to it [Dwyer \[2015\]](#), [Castrén et al. \[2022\]](#). Systemic risk research has dealt with this kind of problem
45 before, using CoVaR, copulas, or network models to trace how stress moves between institutions and markets [López-Espinosa et al. \[2013\]](#), [Andrieş et al. \[2022\]](#), [Silva et al. \[2017\]](#). Vine copulas fit naturally here: instead of collapsing BTC's relationship with equities and the dollar into a single correlation number, they let each pair's tail behaviour be modelled on its own terms — which is what is needed to see where, and how, cryptocurrency stress
50 could travel.

This paper tackles both questions for three assets whose joint dynamics have become increasingly policy-relevant: Bitcoin (BTC), the S&P 500 equity index, and the US Dollar Index (DXY). The 2021–2025 sample provides exceptionally rich regime variation, spanning the 2022 crypto bear market (BTC peak-to-trough: -77%), the Federal Reserve
55 tightening cycle (400 bps over 2022–2023), the FTX exchange collapse (November 2022), the approval of US spot Bitcoin ETFs (January 2024), and the fourth halving (April 2024) — a sequence that makes structural stability an empirical question rather than a maintained assumption [Bouri et al. \[2021\]](#).

The literature offers mixed evidence on the BTC–S&P 500 link. Early work found Bit-

60 coin largely decorrelated from equities and currencies [Dyhrberg \[2016\]](#), [Bouri et al. \[2017\]](#). More recent studies document growing co-movement during stress episodes [Conlon and McGee \[2020\]](#), [Corbet et al. \[2020\]](#), but typically rely on DCC-GARCH or rolling correlations that impose Gaussian dependence and cannot detect asymmetric tail behaviour. Papers applying copulas to cryptocurrency markets [BenSaida \[2023\]](#), [Mensi et al. \[2019\]](#) focus on
65 static or dynamic specifications in isolation, without a formal predictive comparison or structural break analysis.

We close these gaps by combining a static C-vine, a DCC-GARCH benchmark, and a rolling dynamic C-vine, all evaluated on a strictly out-of-sample basis using predictive log-likelihoods and Diebold-Mariano tests. Bai-Perron multiple break tests are applied to
70 the rolling Kendall's τ series to date regime changes without imposing break locations a priori [Bai and Perron \[1998\]](#).

The results yield four main findings. First, the static C-vine uncovers a heterogeneous dependence structure invisible to linear measures: a Survival Gumbel copula for S&P 500–BTC captures upper tail co-movement ($\hat{\tau} = +0.218$), while a Student- t copula for the
75 conditional DXY–BTC pair ($\hat{\nu} \approx 5$, $\hat{\tau} = -0.176$) reveals a symmetric tail channel between Bitcoin and the dollar that operates independently of equity conditions. Second, out-of-sample evaluation shows that the static C-vine dominates both the DCC benchmark and the rolling dynamic vine, providing evidence that parsimony outperforms flexibility. Third, Bai-Perron tests identify three structural breaks per pair; most strikingly, the
80 FTX collapse on November 9, 2022 is recovered automatically as a break in BTC–DXY dependence. Fourth, the S&P 500–BTC correlation remains strictly positive across all four regimes, refuting the diversification narrative for Bitcoin in the post-institutionalisation era.

The remainder of the paper is organised as follows. Section 2 presents the methodology,
85 Section 3 describes the data, Section 4 reports and discusses the empirical results, and Section 5 concludes.

2 Methodology

The analysis proceeds in seven steps: (i) GARCH marginal modelling; (ii) residual extraction and validation; (iii) PIT transformation; (iv) static C-vine estimation; (v) DCC-GARCH
90 estimation; (vi) rolling dynamic C-vine estimation; and (vii) out-of-sample predictive comparison and structural break analysis.

2.1 Marginal Distribution Modelling

Financial return series typically exhibit volatility clustering and leptokurtosis [Mandelbrot \[1963\]](#), [Cont \[2001\]](#). Flexible distributional assumptions for the innovation process —
95 including skewed and heavy-tailed families — are critical for obtaining valid pseudo-observations for copula estimation, as misspecified marginals contaminate dependence inference [Nguyen et al. \[2026\]](#). Let $r_{i,t}$ denote the log-return of asset i ($i = 1, 2, 3$ for S&P 500, BTC, DXY) at time t ($t = 1, \dots, T$). Each return series is modelled as:

$$r_{i,t} = \mu_i + \varepsilon_{i,t}, \quad \varepsilon_{i,t} = \sigma_{i,t} z_{i,t} \quad (1)$$

where μ_i is the conditional mean, $\sigma_{i,t} > 0$ is the conditional standard deviation, and $z_{i,t}$
100 are i.i.d. innovations with zero mean, unit variance, and distribution $F_i(\cdot \mid \boldsymbol{\psi}_i)$. Three variance processes are considered.

Standard GARCH(1,1) [Bollerslev \[1986\]](#):

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2, \quad \alpha_i + \beta_i < 1 \quad (2)$$

Exponential GARCH(1,1) [Nelson \[1991\]](#):

$$\ln(\sigma_{i,t}^2) = \omega_i + \alpha_i z_{i,t-1} + \gamma_i (|z_{i,t-1}| - \mathbb{E}|z|) + \beta_i \ln(\sigma_{i,t-1}^2) \quad (3)$$

A negative γ_i implies negative shocks amplify volatility more than positive shocks of equal size [Nelson \[1991\]](#).

GJR-GARCH(1,1) [Glosten et al. \[1993\]](#):

$$\sigma_{i,t}^2 = \omega_i + (\alpha_i + \gamma_i \mathbf{1}_{[\varepsilon_{i,t-1} < 0]}) \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2, \quad \alpha_i + \frac{1}{2}\gamma_i + \beta_i < 1 \quad (4)$$

The standardised innovations follow either a Student- t distribution [Bollerslev \[1987\]](#) or a skewed Student- t [Fernandez and Steel \[1998\]](#), [Hansen \[1994\]](#). Model selection uses a two-stage grid search over $3 \times 3 \times 2 = 36$ candidate specifications. *Stage 1*: a model is retained only if the Ljung-Box test [Ljung and Box \[1978\]](#) at lag 10 is passed at the 5% level for both standardised residuals and their squares. *Stage 2*: among valid models, the one minimising $\text{BIC}_i = -2\hat{\mathcal{L}}_i + k_i \ln T$ is selected.

Standardised residuals are then transformed via the probability integral transform (PIT):

$$u_{i,t} = F_i(\hat{z}_{i,t} \mid \hat{\nu}_i, \hat{\xi}_i) \quad (5)$$

where under correct specification $u_{i,t} \sim \mathcal{U}[0, 1]$ [Joe \[1997\]](#), verified by a Kolmogorov-Smirnov test.

2.2 Vine Copula Framework

The joint density of an R-vine copula for n variables decomposes as [Dissmann et al. \[2013\]](#):

$$f(x_1, \dots, x_n) = \prod_{k=1}^n f_k(x_k) \cdot \prod_{j=1}^{n-1} \prod_{e \in E_j} c_{j_e, k_e | D_e}(F(x_{j_e} \mid \mathbf{x}_{D_e}), F(x_{k_e} \mid \mathbf{x}_{D_e}); \boldsymbol{\theta}_{j_e, k_e | D_e}) \quad (6)$$

For $n = 3$, the decomposition reduces to two pair-copulas in Tree 1 and one conditional copula in Tree 2. Candidate families include Gaussian, Student- t , Clayton, Gumbel, Frank, Survival Clayton, Survival Gumbel, and their $90^\circ/270^\circ$ rotations. Family selection uses the AIC; the tree structure maximises the sum of absolute Kendall's τ values in Tree 1 [Dissmann et al. \[2013\]](#). Recent work on copula-based conditional quantiles confirms that getting the conditional tail dependence right matters in multivariate copula settings [Hakim and Syuhada \[2024\]](#). The same logic drives CoVaR-type systemic risk indicators, where the copula family chosen changes the estimated spillover between institutions or

markets [López-Espinosa et al. \[2013\]](#), [Andrieş et al. \[2022\]](#). The C-vine used here applies
 125 that idea to three assets, with an explicit conditioning structure instead of a single pair.

2.3 Dynamic Dependence Models

2.3.1 DCC-GARCH

The DCC-GARCH model [Engle \[2002\]](#) parameterises the conditional covariance matrix as $\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$, where $\mathbf{R}_t = \text{diag}(\mathbf{Q}_t)^{-1/2} \mathbf{Q}_t \text{diag}(\mathbf{Q}_t)^{-1/2}$ and:

$$\mathbf{Q}_t = (1 - a - b) \bar{\mathbf{Q}} + a \boldsymbol{\varepsilon}_{t-1} \boldsymbol{\varepsilon}_{t-1}' + b \mathbf{Q}_{t-1}, \quad a + b < 1 \quad (7)$$

130 The condition $a + b < 1$ ensures covariance stationarity. Multivariate GARCH models with non-Gaussian innovations have recently been shown to provide superior fit for bivariate financial time series compared to Gaussian specifications [Kang et al. \[2025\]](#).

2.3.2 Rolling Dynamic C-Vine Copula

The rolling dynamic C-vine is estimated on a moving window of $T_0 = 252$ trading days.
 135 Vine copula-based rolling estimation has recently been shown to enhance multivariate forecasting by capturing complex interdependencies among time series [Manjunatha et al. \[2025\]](#). At each step $t = T_0 + 1, \dots, T$, the vine is re-estimated on $[t - T_0, t - 1]$, yielding $\hat{\boldsymbol{\theta}}_{t-1}$. The one-step-ahead predictive log-likelihood score is:

$$\text{PLS}_t = \sum_{j=1}^{n-1} \sum_{e \in E_j} \log c_{j_e, k_e | D_e} \left(u_{j_e | D_e, t}, u_{k_e | D_e, t}; \hat{\boldsymbol{\theta}}_{j_e, k_e | D_e, t-1} \right) \quad (8)$$

2.4 Model Comparison and Structural Break Tests

140 Equal predictive accuracy is tested using the Diebold-Mariano test [Diebold and Mariano \[1995\]](#) with HAC correction (Newey-West, lag = 5):

$$\text{DM} = \bar{d} / \sqrt{\hat{\sigma}_d^2 / N_{\text{pred}}} \quad (9)$$

where $\bar{d} = N_{\text{pred}}^{-1} \sum_t (\text{PLS}_t^A - \text{PLS}_t^B)$ and $\hat{\sigma}_d^2$ is the HAC variance. Predictive AIC and BIC penalise $-2\mathcal{L}_{\text{pred}}$ by $2\bar{k}$ and $\bar{k} \ln N_{\text{pred}}$, respectively.

To detect regime changes, Bai-Perron [Bai and Perron \[1998, 2003\]](#) multiple structural break tests are applied to the rolling Kendall's τ_t series. The motivation is supported by recent work showing that copula-based Markov chain models detect structural changes in sequential financial data more accurately than independence-based alternatives [Sun et al. \[2025\]](#). The procedure minimises:

$$\text{SSR}(T_1, \dots, T_m) = \sum_{j=0}^m \sum_{t=T_j+1}^{T_{j+1}} (\tau_t - \hat{\mu}_j)^2 \quad (10)$$

with $m_{\text{max}} = 3$ and optimal m by the modified BIC of [Liu et al. \[1997\]](#).

3 Data

The dataset contains daily closing prices for BTC/USD, the S&P 500, and the DXY, from January 4, 2021 to December 31, 2025. After aligning to common trading days, the final sample has $N = 1,254$ observations. Log-returns are $r_{i,t} = 100 \times \ln(P_{i,t}/P_{i,t-1})$. Prices are sourced from Yahoo Finance (retrieved January 2026). All computations use R 4.4 with VineCopula [Schepsmeier et al. \[2023\]](#), rugarch/rmgarch [Ghalanos \[2022\]](#), and strucchange [Zeileis et al. \[2002\]](#). Replication code is available upon request.

The sample spans the full post-COVID institutionalisation cycle, including the 2022 crypto bear market (BTC peak-to-trough: -77%), the Federal Reserve rate hike cycle (400 bps, 2022–2023), the FTX collapse (November 8–11, 2022), the launch of US spot Bitcoin ETFs (January 10, 2024), the fourth Bitcoin halving (April 19, 2024), and renewed macroeconomic uncertainty in early 2025.

Descriptive statistics confirm the stylised facts motivating the GARCH-copula approach. BTC is by far the most volatile series (annualised $\hat{\sigma} \approx 68\%$) vs. S&P 500 ($\approx 19\%$) and DXY ($\approx 7\%$). All three series exhibit excess kurtosis and negative skewness. Augmented Dickey-Fuller tests reject the unit root null at the 1% level. Figure 1 shows the daily log-return series.

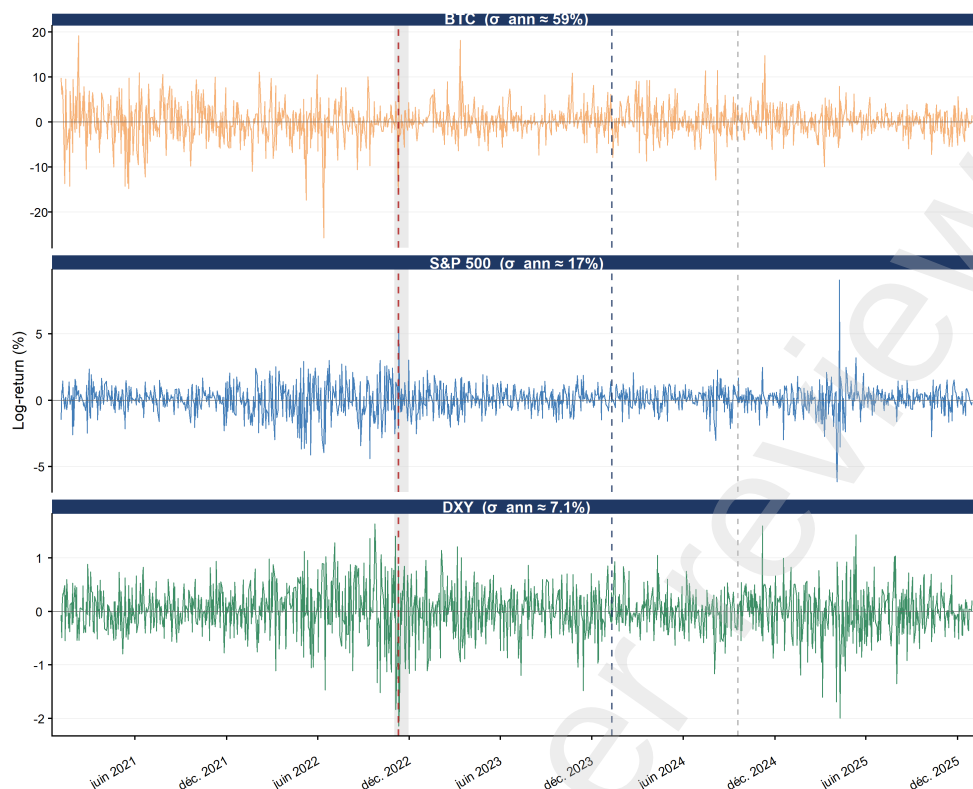


Figure 1: Daily log-returns of BTC, S&P 500, and DXY (January 2021–December 2025). Shaded area: FTX collapse (November 2022). Vertical dashed lines: Bitcoin ETF approval (January 2024) and fourth halving (April 2024).

4 Empirical Results

4.1 Marginal Distribution Models

Table 1 reports the selected GARCH models. S&P 500 is best described by an eGARCH(1,1) with skewed Student- t innovations, consistent with the well-documented negative asymmetry of equity returns. BTC also requires an eGARCH(1,1) but with symmetric Student- t errors: its distribution is heavy-tailed yet sign-symmetric. DXY is captured by a standard sGARCH(1,1) with Student- t errors. All three models pass the Ljung-Box test: p -values on squared residuals are 0.826, 0.731, and 0.713 for S&P 500, BTC, and DXY, respectively. The eGARCH leverage parameter for BTC, $\hat{\gamma}_1 = 0.126$ ($p < 0.001$), indicates a pronounced asymmetric volatility response absent from DXY and weaker in S&P 500.

Table 1: Selected GARCH Models and Diagnostic Tests.

Series	Model	LB(10)	LB ² (10)	BIC
S&P 500	eGARCH(1,1)-ARMA(0,0)-sstd	0.8127	0.8260	2.6547
BTC	eGARCH(1,1)-ARMA(0,0)-std	0.7245	0.7311	5.2668
DXY	sGARCH(1,1)-ARMA(0,0)-std	0.0655	0.7127	1.1390

Note: LB(10) and LB²(10) are Ljung-Box p -values on standardised and squared residuals (lag = 10).
All p -values > 0.05 confirm adequate residual whitening.

4.2 Static Dependence Structure

4.2.1 Kendall's Tau Matrix

Table 2 shows three clear patterns. S&P 500–BTC exhibits a moderate positive rank correlation ($\hat{\tau} = +0.238$), consistent with the growing integration of Bitcoin into equity markets documented in Section 4.5. S&P 500–DXY is negatively associated ($\hat{\tau} = -0.177$), tightening during the Fed rate hike cycle. BTC–DXY shows the weakest connection ($\hat{\tau} = -0.112$), suggesting Bitcoin's response to dollar movements is mainly indirect, transmitted through the equity channel.

Table 2: Empirical Kendall's Tau Matrix.

	S&P 500	BTC	DXY
S&P 500	1.0000	+0.2379	−0.1774
BTC	+0.2379	1.0000	−0.1122
DXY	−0.1774	−0.1122	1.0000

Note: Computed on PIT-transformed GARCH residuals. All off-diagonal elements significant at the 5% level.

4.2.2 C-Vine Copula Estimation

Table 3 reports the static C-vine with S&P 500 as the central node. The C-vine structure has been shown to yield reliable inference even in small-to-moderate samples when combined with parametric bootstrap uncertainty quantification Zhang et al. [2025]. In Tree 1, the S&P 500–BTC pair is modelled by a Survival Gumbel copula ($\hat{\theta} = 1.278$, SE = 0.113, $p < 0.001$, $\hat{\tau} = +0.218$), which places greater probability mass in the upper joint tail — since Bitcoin and equities co-move more strongly in rallies than in downturns, diversi-

fication benefits are reduced precisely when they are most needed. The bootstrap 95% interval $[1.057, 1.500]$ lies entirely above unity (the independence boundary), confirming statistical robustness. The S&P 500–DXY pair follows a Frank copula ($\hat{\theta} = -0.541$, SE = 0.280, $p = 0.053$, $\hat{\tau} = -0.060$), with borderline significance reflecting the gradual decoupling of S&P 500 and DXY since 2020. Moving to Tree 2, the pair DXY–BTC *conditional on* S&P 500 is described by a Student- t copula ($\hat{\rho} = -0.273$, SE = 0.066, $p < 0.001$, $\hat{\tau} = -0.176$), with 95% interval $[-0.402, -0.144]$. The degrees of freedom $\hat{\nu} \approx 5$ indicate symmetric tail dependence between Bitcoin and the dollar once equity conditions are controlled for — a channel invisible to linear correlation. The goodness-of-fit test of White [White \[1982\]](#) confirms adequate overall fit (stat = 6.881, $p = 0.520$).

Table 3: Static C-Vine Copula: Pair-Copula Decomposition with Bootstrap Standard Errors.

Tree	Pair	Family	Par ₁	Par ₂	$\hat{\tau}$
T1	S&P 500–BTC	Surv. Gumbel	1.2784*** (0.1130)	—	+0.2178*** (0.0691)
T1	S&P 500–DXY	Frank	−0.5407† (0.2795)	—	−0.0599† (0.0308)
T2	DXY–BTC S&P 500	Student- t	−0.2728*** (0.0659)	5.1915	−0.1759*** (0.0436)

Bootstrap SEs in parentheses ($B = 500$, structure fixed). $\hat{\tau}$ SE via delta method. *** $p < 0.001$;

† $p = 0.053$. S&P 500 is the central node. GOF [White \[1982\]](#): stat = 6.881, $p = 0.520$. LogLik = 154.20; AIC = −300.40; BIC = −279.87; $k = 4$.

4.3 Dynamic Conditional Correlations

Table 4 reports the DCC-GARCH estimates. The reaction parameter $\hat{a} = 0.014$ ($p = 0.002$) and persistence $\hat{b} = 0.975$ ($p < 0.001$) imply correlation shocks have a half-life of ≈ 27 trading days. The stationarity condition $\hat{a} + \hat{b} = 0.989 < 1$ is satisfied. The S&P 500–DXY and BTC–DXY correlations fluctuate around negative means (−0.221 and −0.128) and occasionally change sign, reflecting regime-dependent dollar dynamics. By contrast, the S&P 500–BTC correlation remains strictly positive throughout the sample (mean = +0.338, range $[+0.079, +0.538]$), providing no support for a diversification role of Bitcoin.

Figure 2 traces the full time path.

Table 4: DCC-GARCH Results.

(a) Joint DCC Parameters.

Parameter	Estimate	Std. Error	t -stat	p -value
dcca1 (a)	0.0143	0.0047	3.047	0.002
dccb1 (b)	0.9749	0.0100	97.77	0.000
$a + b$	0.9892	—	—	< 1

(b) Dynamic Conditional Correlations — Summary Statistics.

Pair	Mean $\hat{\rho}(t)$	Min $\hat{\rho}(t)$	Max $\hat{\rho}(t)$
S&P 500–BTC	+0.3383	+0.0790	+0.5381
S&P 500–DXY	−0.2210	−0.4857	+0.1563
BTC–DXY	−0.1280	−0.3617	+0.1468

Note: Student- t DCC estimated by two-step QML. Half-life ≈ 27 days ($\ln 0.5 / \ln \hat{b}$). $\hat{a} + \hat{b} < 1$ ensures covariance stationarity.

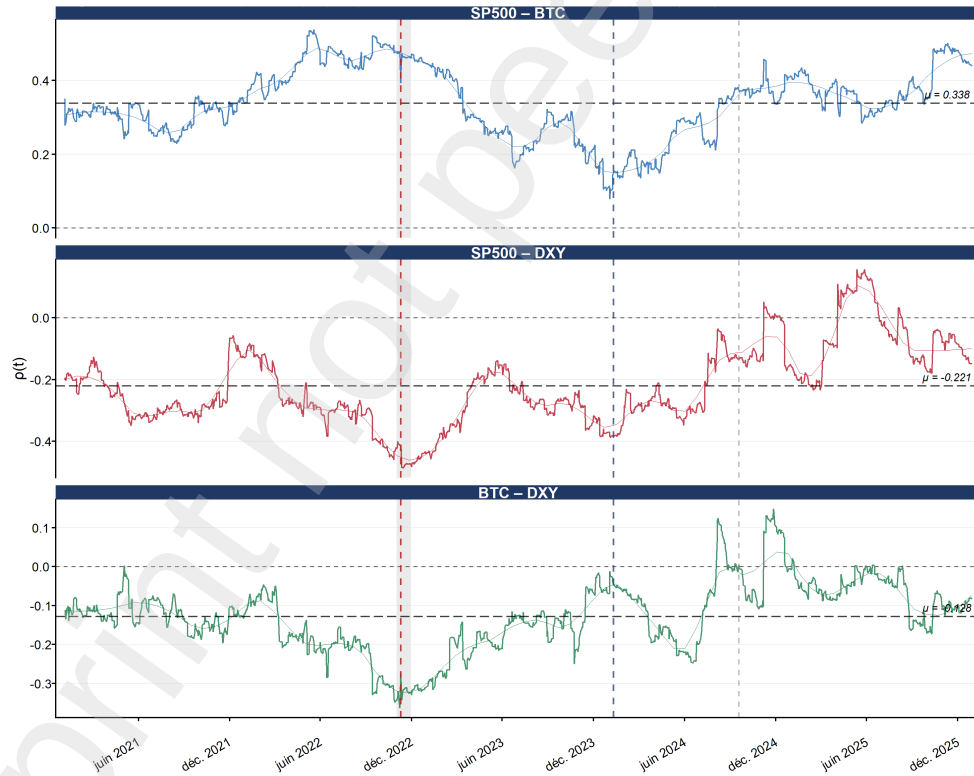


Figure 2: Dynamic Conditional Correlations from DCC-GARCH(1,1), January 2021–December 2025. Shaded regions: FTX collapse (November 2022), Bitcoin ETF approval (January 2024), fourth halving (April 2024). Dashed lines: sample means.

4.4 Out-of-Sample Predictive Comparison

Table 5 compares predictive log-likelihoods over $N_{\text{pred}} = 1,002$ observations. The static C-vine ranks first (log-score 0.1366/obs), ahead of the DCC-Student- t model (0.1301) and the rolling dynamic vine (0.1236). The vine's advantage over the DCC benchmark reflects its ability to capture pair-specific and asymmetric tail structures — upper-tail concentration for S&P 500–BTC via the Survival Gumbel, and symmetric conditional tails for DXY–BTC via the Student- t copula — that the scalar correlation parameterisation of the DCC cannot accommodate. The dynamic vine ranks last: no single copula family exceeds 32% of rolling windows, so frequent family switching adds estimation noise that outweighs any gain from time variation. Diebold-Mariano p -values (0.522, 0.195, 0.542) do not reject equal accuracy, but the consistent ordinal ranking across all criteria supports the conclusion that parsimony dominates Patton [2012], Christoffersen [2012].

Table 5: Out-of-Sample Predictive Comparison ($N_{\text{pred}} = 1,002$).

(a) Log-Predictive Score Comparison.

Model	LogLik	LogLik/obs	AIC _{pred}	BIC _{pred}
Static C-Vine	136.87	0.1366	−265.74	−254.02
DCC (Student- t)	130.39	0.1301	−256.78	−247.50
Dynamic C-Vine	123.89	0.1236	−240.55	−222.77

(b) Diebold-Mariano Tests (HAC, Newey-West lag = 5).

Comparison	DM stat	p -value
Static vs. DCC-Student- t	0.641	0.522
Static vs. Dynamic	1.295	0.195
DCC-Student- t vs. Dynamic	0.610	0.542

Note: H_0 = equal predictive accuracy. Estimation window $T_0 = 252$ days.

4.5 Structural Break Analysis and Copula Family Stability

4.5.1 Bai-Perron Breakpoint Tests

Tables 6 and 7 report the detected break dates and corresponding segment means $\hat{\mu}_j$ of rolling Kendall's $\hat{\tau}_t$. Each pair yields four economically interpretable regimes.

The most striking result concerns the BTC–DXY pair. The first break falls precisely on

November 9, 2022 — the day the FTX collapse became public — recovered automatically without any user-imposed constraint. The segment mean shifts from $\hat{\mu}_1 = -0.189$ to $\hat{\mu}_2 = -0.073$ ($> 60\%$ reduction in absolute dependence), consistent with a flight-to-quality episode in which the dollar strengthened while Bitcoin sold off sharply. A second shift in late November 2023 ($\hat{\mu}_3 = -0.114$) coincides with ETF approval anticipation, and a third in April 2025 ($\hat{\mu}_4 = -0.047$) suggests progressive normalisation.

For BTC-S&P 500, mean $\hat{\tau}$ rises near-monotonically from +0.071 (segment 1) to +0.261 (segment 3), documenting deepening Bitcoin-equity integration, before a mild pullback to +0.198 in segment 4 following post-halving normalisation. For DXY-S&P 500, dependence peaks in absolute terms at segment 2 ($\hat{\mu}_2 = -0.301$, rate hike cycle) before attenuating to $\hat{\mu}_4 = -0.098$ as the Fed pivoted toward cuts. Across all pairs, the segment-level $\hat{\tau}$ values align closely with known market events, confirming genuine regime transitions. Figure 3 illustrates the rolling series with break dates and segment means superimposed.

Table 6: Bai-Perron Multiple Breakpoint Tests on Rolling Kendall's τ .

Pair	Break 1	Break 2	Break 3
BTC-S&P 500	2023-02-22	2023-12-08	2024-07-17
DXY-S&P 500	2023-12-19	2024-07-26	2025-03-04
DXY-BTC	2022-11-09*	2023-11-30	2025-04-15

*FTX collapse (November 8–11, 2022). Rolling window = 252 days. Max 3 breaks. Break selection via the modified BIC of Liu et al. [1997].

Table 7: Segment Mean Kendall's $\hat{\tau}$ Across Bai-Perron Regimes.

Pair	Segment 1	Segment 2	Segment 3	Segment 4
BTC-S&P 500	+0.071	+0.142	+0.261	+0.198
DXY-S&P 500	-0.183	-0.301	-0.214	-0.098
DXY-BTC	-0.189	-0.073	-0.114	-0.047

Note: Segment boundaries from Table 6. Values are means of rolling $\hat{\tau}_t$ within each segment. Near-monotone increase in BTC-S&P 500 across segments 1–3 documents progressive Bitcoin-equity integration. Attenuation of DXY-BTC in segment 2 coincides with FTX collapse.

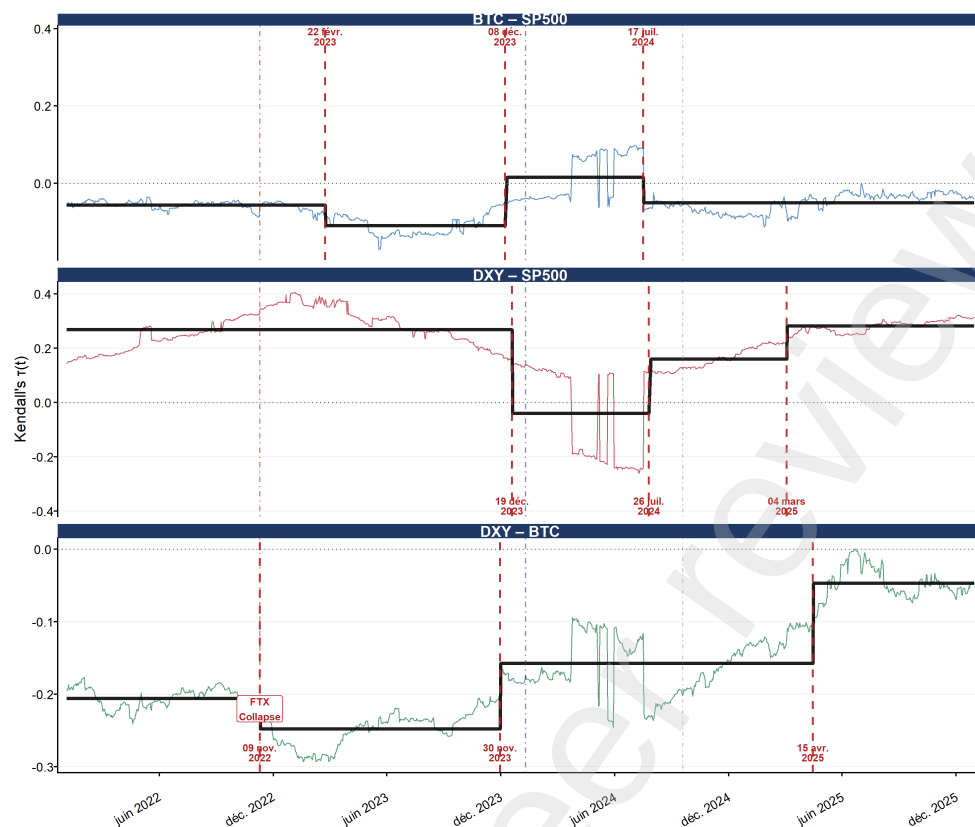


Figure 3: Rolling Kendall's τ and Bai-Perron Structural Breaks (window = 252 days). Solid lines: rolling τ . Thick lines: segment means. Vertical red dashes: break dates. Vertical grey dashes: key market events (FTX collapse, ETF approval, Fed cut).

4.5.2 Rolling Copula Family Stability

Table 8 reports the two most selected copula families across 1,002 rolling windows. The marginal pairs exhibit pronounced instability: for BTC-S&P 500, Frank leads at 31.9% but no family exceeds one third of windows; the second choice, rotated Joe 270° (21.4%), captures a qualitatively different tail structure. The DXY-S&P 500 pair is even more indeterminate (Gaussian 27.2%, Frank 26.9%). By contrast, the conditional DXY-BTC pair is substantially more stable: Student- t is selected in 42.0% of windows, nearly double the second family (Frank 21.6%), consistent with the static C-vine. This contrast between marginal instability and conditional stability clarifies the predictive ranking from Section 4.4: the vine's informational advantage resides in the conditional pair, whose structure is robust enough to survive time aggregation.

Table 8: Dominant Copula Families in Rolling C-Vine ($N = 1,002$ Windows).

Pair	1st Family	Freq. (%)	2nd Family	Freq. (%)
BTC–S&P 500	Frank	31.9	Joe270	21.4
DXY–S&P 500	Gaussian	27.2	Frank	26.9
DXY–BTC	Student- t	42.0	Frank	21.6

Note: Estimation window = 252 days. Family selected by AIC. DXY–BTC conditional pair shows the highest stability.

5 Conclusion

This paper studied the static and dynamic dependence among Bitcoin, the S&P 500, and the DXY over January 2021 to December 2025, combining GARCH-filtered marginals, a static C-vine copula, a DCC-GARCH model, and a rolling dynamic C-vine evaluated on a strictly out-of-sample basis. Three main findings stand out.

First, the static C-vine uncovers a heterogeneous dependence structure across pairs and trees. The Survival Gumbel for S&P 500–BTC captures upper tail co-movement (Bitcoin and equities cluster more in rallies than in downturns), limiting diversification when it is most needed. The Student- t for DXY–BTC given S&P 500 ($\hat{\nu} \approx 5$) reveals a symmetric tail channel between Bitcoin and the dollar invisible to linear correlations. Bootstrap inference confirms robustness: the 95% interval for the Survival Gumbel parameter $[1.057, 1.500]$ lies entirely above the independence boundary.

Second, the static C-vine predicts best out-of-sample (0.1366/obs), ahead of both the DCC-Student- t and the dynamic vine. The dynamic vine’s poorer performance reflects copula family instability across rolling windows: flexibility increases noise rather than accuracy [Patton \[2012\]](#).

Third, Bai-Perron tests recover the FTX collapse of November 9, 2022 automatically as a break in BTC–DXY dependence, and the four identified regimes provide a statistically grounded timeline of Bitcoin’s market integration.

Two results here matter specifically for financial stability monitoring. The upper tail co-movement between BTC and equities, combined with the sharp break around the FTX collapse, shows that crypto markets can transmit stress into traditional markets,

not just absorb it [Dwyer \[2015\]](#), [Castrén et al. \[2022\]](#). Practically, this cuts two ways.

275 Standard CoVaR-type indicators built on linear or unconditional co-movement [López-Espinosa et al. \[2013\]](#), [Andrieş et al. \[2022\]](#) will underestimate joint downside risk here, because the dependence we find is asymmetric and concentrated in the tails rather than uniform. And because that dependence moved substantially around the FTX event, a systemic risk indicator built on crypto-traditional linkages probably needs a rolling or
280 regime-aware window rather than a single full-sample estimate — co-movement between markets tends to spike during crises, not stay constant [Mobarek et al. \[2016\]](#).

From a portfolio standpoint, managers should also note that upper tail dependence limits Bitcoin’s diversification value during bull markets. Risk managers relying on Gaussian copulas will underestimate tail risk: a preliminary Monte Carlo exercise suggests
285 an Expected Shortfall gap of $\approx 6.8\%$ at the 99% level for a 50/50 BTC–S&P 500 portfolio, though formal validation using ES-specific backtests [Acerbi and Székely \[2014\]](#) is left for future work. Future research could extend the framework to wider asset universes, use observation-driven vine models with fixed family structure, or incorporate macroeconomic state variables (VIX, yield curve slope) to better understand what drives the
290 detected regime changes, as well as embed the estimated pairwise dependence structures into a formal systemic risk indicator (e.g. a crypto-augmented CoVaR) to quantify directly the contribution of cryptocurrency stress to downside risk in traditional markets.

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300 The authors report there are no competing interests to declare.

Data Availability Statement

Daily closing prices were obtained from Yahoo Finance (January 2026) and are available from the corresponding author upon request. Replication code in R is also available upon request.

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